

AI-Powered Tourism Platform — Project Guide & Architecture Summary

A Complete, Simple Explanation of How the Entire System Works Under the Hood

Project: TravelNepal (Minor Project)	Stack: Next.js 16 + FastAPI + LangGraph
Database: PostgreSQL (Neon DB) + Drizzle ORM	Payment & Media: Khalti / eSewa + Cloudinary
AI Engine: Google Gemini 2.5 + Edge TTS Voice	Emergency: 1-Click SOS with Offline SMS Relay

1. Executive Summary — What is TravelNepal?

TravelNepal is an all-in-one digital tourism ecosystem built for travelers exploring Nepal and local tourism service providers (Hotels, Restaurants, Tour Guides). Unlike basic static travel directories, TravelNepal merges an **autonomous multi-agent AI voice assistant**, **verified live database grounding**, **Human-In-The-Loop (HITL) safe action execution**, **role-based business workspaces**, **natural voice expense tracking**, and a **zero-internet 1-Click SOS emergency protocol**.

In simple terms: A traveler can talk or type to plan a trip, get verified hotel and food recommendations, book rooms using Khalti, log daily expenses via voice, and trigger emergency alerts in remote mountain trails without internet access.

2. High-Level System Architecture

The application is structured into three clean, interconnected tiers:

Frontend Tier (Next.js 16)	Full-stack Next.js App Router with TypeScript, Tailwind CSS, Shadcn UI components, Web Speech API (continuous voice), and Server Actions.
AI & Agentic Backend (FastAPI + LangGraph)	FastAPI microservice hosting a LangGraph state graph. Coordinates specialized Gemini 2.5 sub-agents (Supervisor, Hotel, Restaurant, Itinerary, Expense).
Database & Cloud Infrastructure	PostgreSQL database with Drizzle ORM schemas (Users, Hotels, Rooms, Restaurants, Menus, Guides, Bookings, Expenses, SOS Alerts) + Cloudinary CDN.

3. Core Features Explained in Simple Terms

A. Autonomous Multi-Agent AI & Voice Assistant

When a user asks: *'Find verified hotels near Dharan for 4 people'* or speaks into the microphone:

- **Continuous Speech Recognition:** Web Speech API listens to the user. A 3-second silence debounce automatically finishes the speech input without cutting the user off.

- **LangGraph Supervisor Routing:** The FastAPI backend passes the query to a *Supervisor Agent* that classifies intent (Hotel Search, Restaurant Discovery, Trip Planning, Expense Logging, or Booking).
- **Direct Database Grounding:** The specialized agent queries the live PostgreSQL database for real registered listings with accurate prices and IDs.
- **Actionable Cards & Voice Narration:** The response renders rich UI cards linking directly to `/hotels/[id]`, Google Maps, and plays natural studio audio voice guidance.

B. Human-In-The-Loop (HITL) Safety Proposals

Instead of allowing AI to write blindly to the database, any state-changing action triggers an interactive **Action Proposal Card**:

- **Room & Dish Additions:** Hotel and Restaurant owners can add rooms or menu items with editable form fields (Price in NPR, prep time, room type) and Cloudinary photo upload.
- **Booking Proposals:** When booking a stay, the AI presents a confirmation card with guest count, dates, and total amount before executing the payment.
- **User Control:** The action is only executed when the user clicks 'Confirm & Execute'.

C. Multi-Role Workspaces & Role-Based Access (RBAC)

TravelNepal provides specialized dashboards based on authenticated user roles:

- **Traveler:** Destination discovery, interactive itinerary planner, booking history, expense ledger, 1-click SOS.
- **Hotel Owner:** Room inventory management, nightly rate configuration, guest reservation tracking, verification status.
- **Restaurant Owner:** Digital food menu management, dish categories, opening/closing switch, dining orders.
- **Tour Guide:** Trekking tour packages, duration, pricing, verified license credentials.
- **Super Admin:** Partner KYC identity reviews, platform statistics, company/destination directory, chronological activity alerts.

D. Voice-Powered Travel Expense Tracker

Travelers can manage their trip finances by speaking naturally, e.g. *'Add 20 rupees in expense for a drink'*:

- **Smart NLU Parsing:** The regex & LLM parser extracts the amount (NPR 20), item name (Drink), and automatically classifies the category (Food & Dining).
- **HITL Proposal:** Prompts an inline confirmation card before logging the transaction to the database.
- **Analytics Dashboard:** Aggregates category spending (Accommodation, Transport, Food, Activities) with visual breakdown charts.

E. 1-Click SOS Emergency Panic System (Offline Resilient)

Safety is paramount for Himalayan trekkers. The `/emergency` page features:

- **One-Click Panic Trigger:** Instantly acquires device GPS coordinates and plays an emergency warning tone.
- **Zero-Internet SMS Relay:** Automatically pre-composes and launches a cellular SMS with live Google Maps coordinates to Tourist Police (1144), Nepal Police (100), or custom emergency contacts.

- **Toll-Free Hotline Directory:** 1-tap phone dialers for Flood Rescue (1155), Red Cross Ambulance (102), and Mountain Helicopter Rescue (+977 1-4444555).

4. Digital Payment & Reservation Lifecycle

Step	Event / Action	Technical Mechanism
1	Traveler Selects Room / Tour	Picks dates & guests on /hotels/[id] or via AI Chat proposal.
2	Khalti / eSewa Gateway Initiated	Server creates pending booking record and requests payment URL.
3	Digital Wallet Authorization	User approves payment in Khalti/eSewa secure sandbox.
4	Instant Verification Callback	Webhook verifies transaction ID and confirms booking status.
5	Workspace Notification Dispatched	Hotel owner and Super Admin receive instant chronological alerts.

5. Presentation & Viva Q&A; Guide

Here are simple, high-scoring answers to common questions your examiners or audience may ask:

Q1: What is the main innovation of your project compared to platforms like Booking.com or TripAdvisor?

A: TravelNepal integrates autonomous multi-agent AI (LangGraph) for natural voice-driven planning with verified database grounding, provides localized role-based partner workspaces with Khalti payments, and includes a zero-internet 1-Click SOS emergency SMS protocol tailored for Nepal's remote trekking geography.

Q2: How does the AI avoid hallucinations when recommending hotels or prices?

A: We use Retrieval-Augmented Generation (RAG) and direct PostgreSQL catalog grounding. The AI query node executes structured database queries for verified hotel/room records and outputs strict entity IDs and links (`/hotels/[id]`).

Q3: What is Human-In-The-Loop (HITL) and why is it implemented?

A: HITL ensures the AI never performs destructive or financial database writes autonomously. When an action is detected (e.g. adding a room, booking a stay, or logging an expense), the agent displays an editable proposal card that requires explicit user confirmation.

Q4: How does the SOS system work if the user has no internet on a mountain trek?

A: It uses the browser Geolocation API to fetch device GPS coordinates offline and automatically opens the native cellular SMS app with a pre-formatted distress message addressed to Tourist Police (1144) over standard 2G/GSM cellular bands.

Q5: Which database and ORM are used, and how are roles handled?

A: We use PostgreSQL on Neon DB with Drizzle ORM for type-safe schema migrations. User authentication is powered by NextAuth.js with multi-role access control (Traveler, Hotel Owner, Restaurant Owner, Tour Guide, Admin) requiring Admin KYC approval.